POLYCYSTIC OVARY SYNDROME (PCOS)

1st described by Stein and leventhal (1935)

Causes: Theories:

Old theory: 1-↓ FSH – receptors on ovaries.

2- ↓ aromatase enzyme.

3- Thick tunica albuginea.

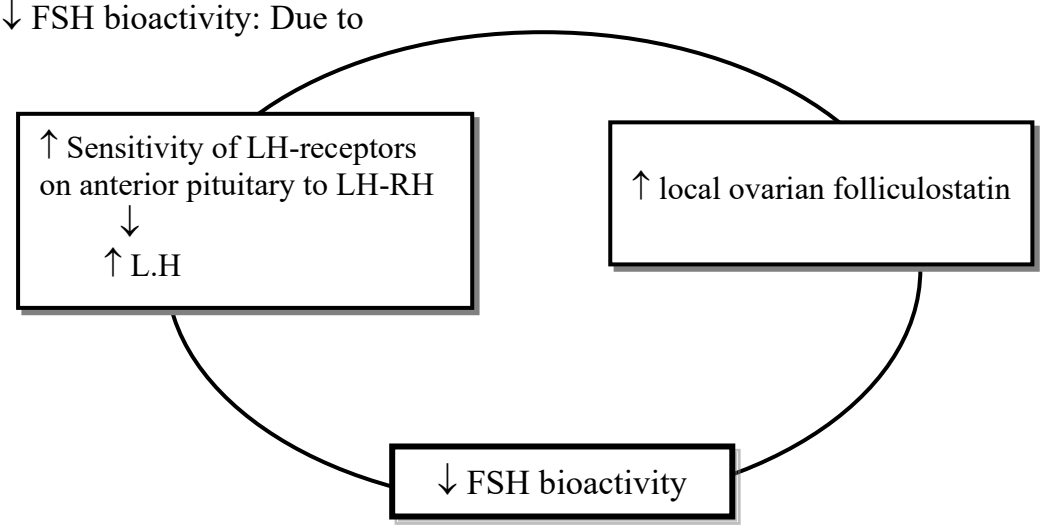
Recent theory:

1- Genetic theory: X-linked dominant.

2-Role of insulin, some sort of insulin resistance which leads to hyper insulinemia which leads to :

- ↑ LH, ↓ FSH
- Decrease liver SHBG.
- Increase adrenal cortex androgen.
- Increase ovarian IGF-1.

3- ↓ FSH bioactivity: Due to



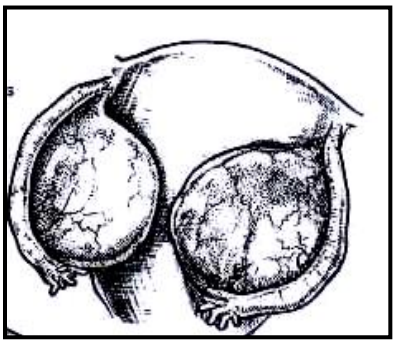
↑ LH leads to:

- ↓ FSH bioactivity → anovulation.
- Ovarian hyperthecosis with ↑ ovarian Androgens → hirsutism.
- ↑ peripheral conversion of androgen → estrone → amenorrhea.

4- Centrally decrease dopamine.

5- Centrally increase ACTH.

C/P:



1- Anovulation and infertility.

2- Oligohypomenorrhea or even amenorrhea.

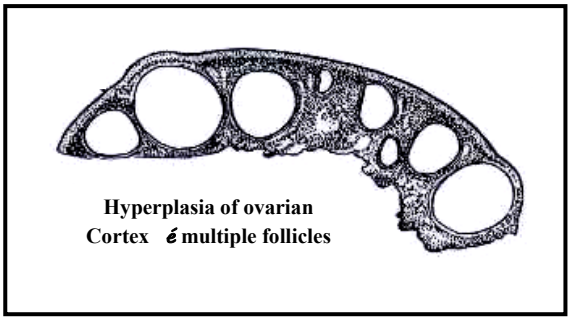
3- Hirsutism.

4- Associated with obesity and hypertension & D.M.

5- Acanthosis nigricans: hyper- pigmented velvety areas on the nape, axillae, inner thigh and vulva.

Investigation:

(A) Biochemical



1- LH: FSH 3:1

2- ↑ androgen (testesterone, androstendione, DHEA, DHEA-S).

3- ↑ (E) especially (E_1).

4- ↑ insulin level (N : 12 – 14 μIU/ml).

(B)U/S:

1- Especially transvaginal U/S shows necklace appearance.

2- 3-D U/S.

3- Doppler : increase blood flow to the ovary.

(C) Laparoscopy:

- Bilateral enlarged ovaries .
- Smooth surface (oyster-like).

(D) Tests for ovulation.

✍ **N.B.: Diagnosis of PCOS based on ESHRE, 2003, 2 of 3:**

(1) Oligo or anovulation.

(2) Clinical and / or lab detection of increase androgen.

(3) U/S of the ovary:

- > 10 peripheral occurring follicle.
- Diameter 2 – 9 mm (< 10 mm).
- Increase ovarian volume (> 10 ml)

} Even in one or both ovaries:

✍ **N.B. Role of obesity and PCOS**

⇨ BMI $\frac{\text{Weight Kg}}{\text{Length m}^2}$ N = 20 – 25

25 – 30 obesity.

> 30 extreme obesity.

⇨ Increase orexogenic factors:

- Increase leptin, adiponection
- Increase central neuropeptide - Y, galanin
- Increase Gherlin.

⇨ Obesity → Increased fasting level of insulin:

- IGF-1 → Increase ovarian androgens.
- Decrease SHBG → Increase free testosterone

⇨ Peripheral conversion of A → E1:

- C/P and complications of hyperestrogenism.
- High flat non oscillating → (Increase LH, Decrease FSH, Increase PRL).

⇨ Obesity → hypertension.

D.D. : PCO – like syndromes = D.D. of hirsutism + oligomenorrhea.

	PCO	**CAH**	**Cushing**	**Androgen secreting tumour**	**Hyperthecosis**
* Time of onset.	15-25 y	Congenital	Anytime	Anytime	Anytime
* Virilization	Rare	(+ve)	Unusual	+ve	+ve
* Testosterone	↑	↑↑	N/↑	↑↑	↑↑
* 17-OHP	N / ↑	↑↑	N / ↑	N / ↑	N / ↑
*Cortisol post-Dex	Suppressed	Suppressed	Not suppressed	Ovarian suppress Adrenal : not suppress	Suppressed
* LH	↑	N.	N.	↓	N.
* FSH	↓	N.	N.	↓	N.

Treatment:

I- General Lines:

1- Decrease in body weight is mandatory before any treatment strategy.

2- Decrease smoking as it increase adrenal androgen secretion.

3- Control hypertension and screening for D.M.

II- Medical :

1- Induction of ovulation:

Q- problems of ovulation indication in PCOS?

(1) Increase PRL → Add dopamine agonist (Bromocryptine, lissuride).

(2) Increase ACTH → Add dexazone 0.5 mg tablet at bed night for 10 days.

(3) Increase insulin → (hyperensulinemia) treated by:

- Increase sensitivity of receptors.
  - * Metformine HCL 500 mg.
  - * Pioglitazone tablet (Glustin) or Rosiglitazone (Avandia).
- Decrease Pancreatic Opiates by naltrexone (Rivia 50 mg tablet).
- Decrease IGF-1 by long acting somatostatin analogue (Octreotide).

(4) Increase androgen treated by wedge resection or ovarian drilling.

(5) obesity treated by reduction of weight.

(6) LH hypersecretion treated by:

- Prestimulatory suppression by:
  - * GnRh – agonist continuously for 6-8 weeks.
  - * COCP for 2-3 months.
  - * Progesterone 5 days before stimulation.
  - * Naltrexone decrease peripheral secretion of insulin leads to decrease of LH.
- Use of low dose set up technique or set down technique.
- Preparation with no LH – like activity e.g. puregon.

2- Treatment of hirsutism: by antiandrogenic drugs.

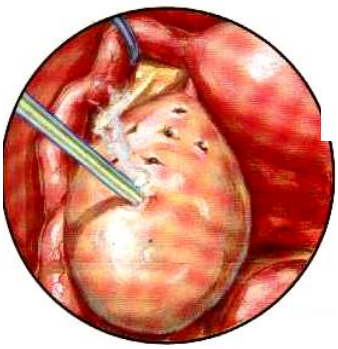
3- Treatment of oligohypomenorrhea: by OCP to regulate the cycle.

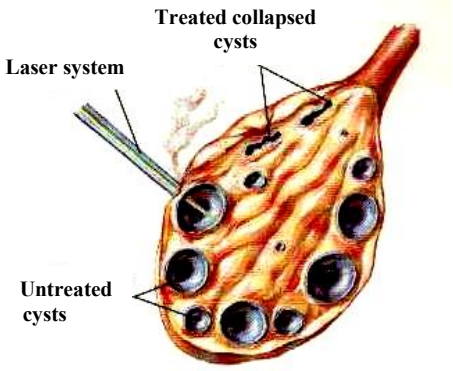
III- Surgical :

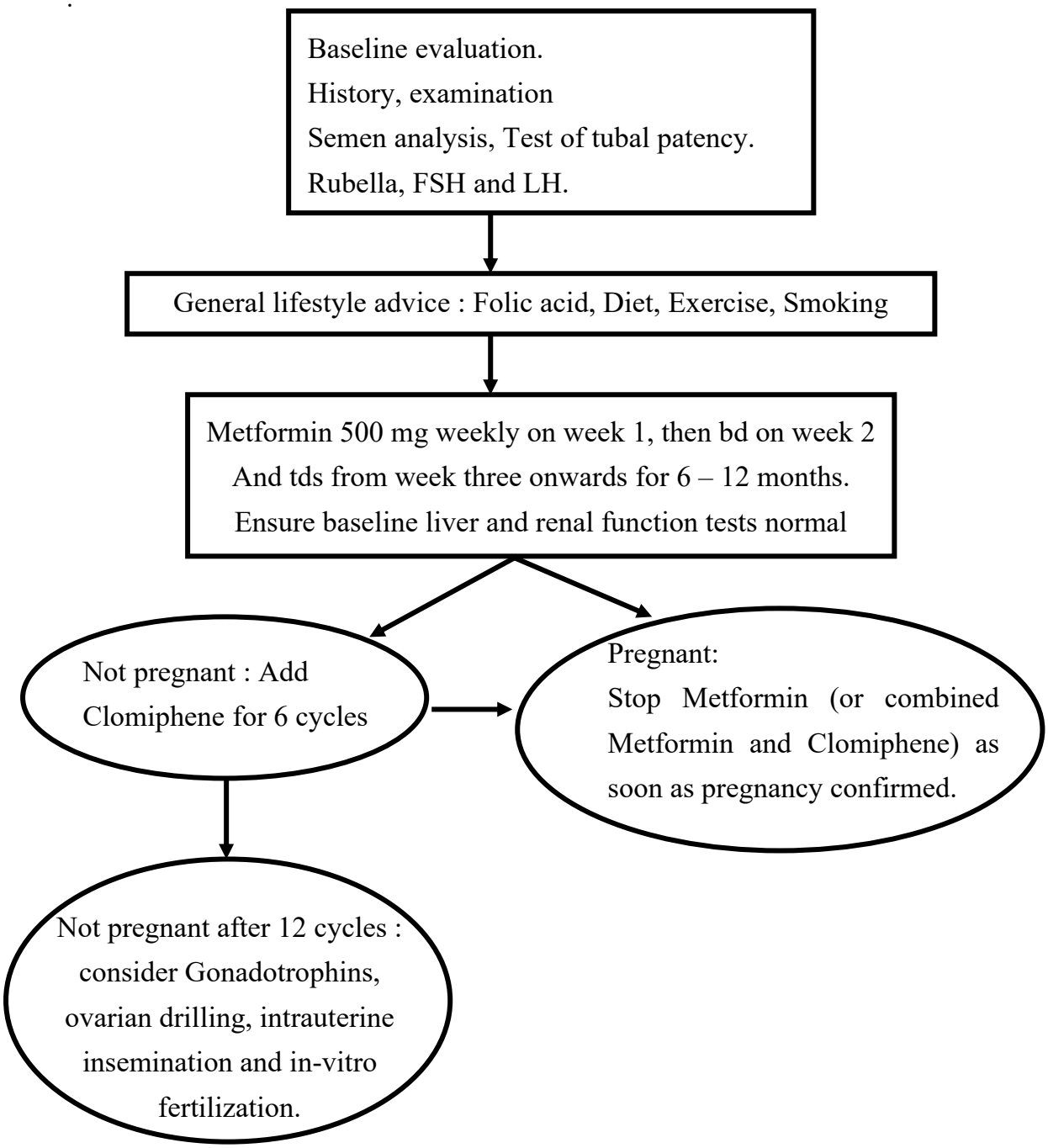
1- Mini-lap: wedge resection:

* Mechanism of action.
  - Decrease stroma → decrease androgens → decrease E (both inhibit ovulation). → (withdrawal bleeding following the operation is a good sign).
  - Ovarian surgical trauma may correct the error and stimulate normal function.
* How much to remove:
  - Remove amount sufficient enough to leave the ovary in normal size.
* Types of wedges removed:
  - Classic wedge remove wedge its base is at the ovary surface + good stromal bulk.
  - Inverted wedge = it's a pex at the surface and its base at ovary stroma, has advantage that main bulk removed is from stroma → decrease androgens + preserving cortical tissue as possible.

2- Laparoscopy: → drill → diathermy
→ endcoagulator
→ laser







Algorithm for the management of infertility in women with PCOS with metformin.